

Neutron histories in the ^3He target

Are scattering, slowing-down and capture in the gas cell included — and does the efficient in-gas moderation enhance the excitation rate?

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14 June 2026

Internal note — informal. Drafted by Claude (Anthropic AI assistant) under D. Neff’s direction; simulation, numbers and text reviewed by D. Neff.

The question

Does the simulation take the full neutron history inside the gas cell — elastic scattering, slowing-down, absorption, capture — into account? And since slowing-down in ^3He is per-collision extremely efficient, could it *enhance* the γ/IPC production by feeding more neutrons into the high-cross-section region? In short: are we at risk of under-counting the number of $^4\text{He}^*$ excitations?

The answer

Yes, the full neutron history is simulated — with high-precision (G4NDL/HP) data and no neutron tracking cut — so the efficient in-gas moderation is included automatically. Counting $^4\text{He}^*$ excitations directly through the dominant $^3\text{He}(\text{n,p})\text{t}$ channel (which is, to a part in 10^4 , the *total* $^4\text{He}^*$ production), the full-transport rate per energy decade lands *at or just below* a naive single-pass estimate. **So we are not under-counting.** The moderation mechanism is real — a single $\text{n}-^3\text{He}$ elastic collision can shed up to 75 % of the neutron energy — but in this thin, 2 cm-bore gas column the elastic mean free path (≈ 24 cm) is far longer than the gas path (≈ 6 cm), so only ~ 20 % of fast neutrons scatter in the gas and the integrated boost to the capture rate is a tens-of-percent perturbation, all of it already inside the simulated numbers. The answer differs by energy regime — opaque at thermal, thin at MeV — but *the scattering physics is switched on identically in every case.*

The rest of this note backs that up: §1 what the simulation actually transports, §2 the excitation budget that shows we are not under-counting, §3 the size of the moderation effect, and §4 event displays that make the three regimes visible.

1 What the simulation transports

Neutrons are not represented by a “capture probability”. They are launched as real particles sampled from the evaluated `n_TOF EAR2` flux (energy *and* energy-dependent transverse profile) upstream of the vessel, and tracked step-by-step through the STEP-derived capsule (pure ^3He at 62.7 mg/cm^3 , 500 bar, in the Al vessel with the 0.9 mm CFRP wrap) and the full detector until they are absorbed or leave the world. Two choices in the physics list (`src/PhysicsList.cc`) matter here:

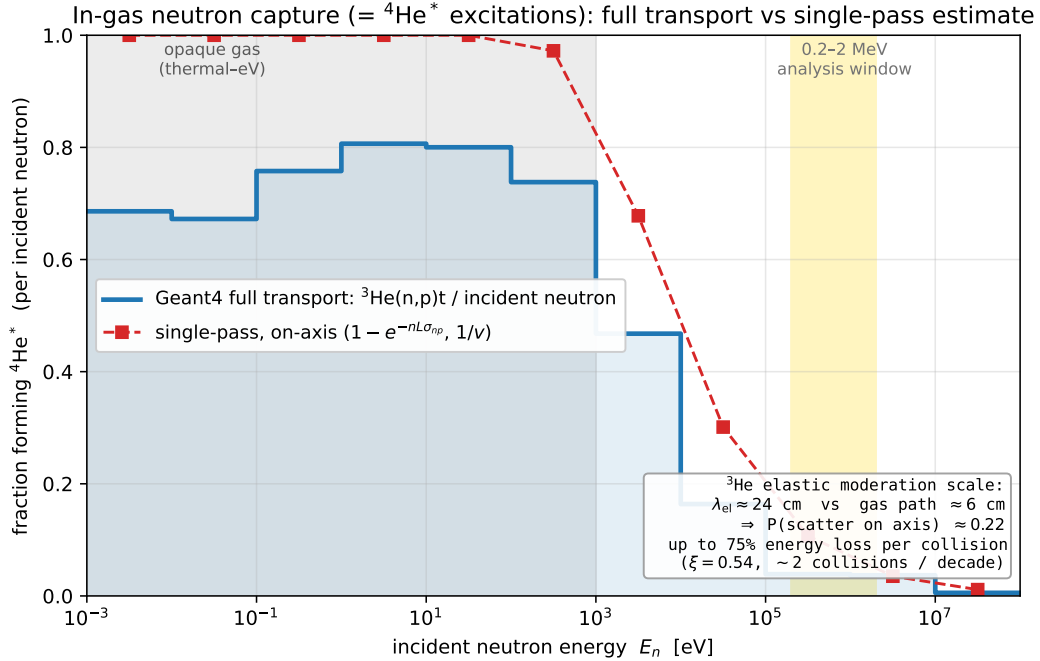
- **High-precision (HP) data are on.** Elastic scattering (the slowing-down), the $(\text{n,p})\text{t}$ channel and radiative capture all run on evaluated G4NDL cross sections with the correct

energy dependence and resonances. Self-shielding, multiple scattering and in-gas moderation are therefore *automatic*, not assumed.

- **The neutron tracking cut is removed on purpose.** Its default $10\ \mu\text{s}$ time limit would silently kill slow neutrons mid-flight (a $0.4\ \text{eV}$ neutron needs $\sim 20\ \mu\text{s}$ to cross $20\ \text{cm}$), destroying exactly the thermal/epithermal transport this experiment depends on. HP is allowed to track neutrons all the way to thermalisation.

2 The excitation budget: are we under-counting?

Every ${}^3\text{He} + n \rightarrow {}^4\text{He}^*$ event is one excitation. Since ${}^4\text{He}^*$ de-excites overwhelmingly through (n,p)t (the γ /IPC/X17 branch is 10^{-8} – 10^{-4} of captures), **the number of excitations is, to a part in 10^4 , just the number of ${}^3\text{He}(\text{n,p})\text{t}$ events** — and that channel has enormous statistics, so the excitation rate can be read straight off the simulation with no branching extrapolation.



From the 5×10^8 -neutron full-range run, the fraction of incident beam neutrons forming ${}^4\text{He}^*$ per energy decade is the blue curve. The answer genuinely differs by regime:

- **thermal $\rightarrow$ $\sim$ keV (opaque):** 67–81 % of beam neutrons that reach the bore are absorbed. The plateau sits below 100 % only because the beam is wider than the 2 cm bore (wider still at low energy) — a geometry effect the simulation gets for free.
- **$\sim 10\ \text{keV} \rightarrow \text{MeV}$ (thinning):** the (n,p)t fraction falls from ~ 0.5 to ~ 0.04 . This is the regime carrying the physics case (0.2–2 MeV window): 95 % of the 714 *directly observed* radiative captures sit above 100 keV.
- **full transport vs. single pass (red dashed):** the naive on-axis $1 - e^{-nL\sigma_{np}}$ ($1/v$) estimate tracks the simulation but sits *above* it at every decade (ratio 0.5–0.8). The deficit is not a missing enhancement — it is the beam-averaged chord being shorter than the 6 cm on-axis path because most of the beam is off-axis. The full sim, with all scattering included, lands at or below the simplest hand estimate, so **we are not silently under-counting**; a careless on-axis thin-target number would if anything *over-count*.

3 Size of the moderation effect

The per-collision intuition is exactly right: ^3He is a light nucleus, so a single elastic $n\text{--}^3\text{He}$ collision can remove up to $f_{\text{max}} = 1 - \left(\frac{A-1}{A+1}\right)^2 = 0.75$ of the neutron energy, with mean log-energy decrement $\xi = 0.54$ — only ~ 2 collisions to drop a decade. Per scatter, ^3He is about as efficient a moderator as exists.

What caps the *integrated* effect is geometry, not efficiency: the elastic mean free path is $\lambda_{\text{el}} = 1/(n\sigma_{\text{el}}) \approx 24\text{ cm}$ ($\sigma_{\text{el}} \approx 3.3\text{ b}$) against a $\sim 6\text{ cm}$ on-axis gas path, so only $\sim 22\%$ of fast neutrons scatter even once in the gas (fewer off axis). At MeV energies roughly a fifth of neutrons get a single large energy kick toward the higher- σ_{np} region and a “second chance” to capture; the rest pass straight through. A real, positive contribution to the rate — fully inside the Geant4 numbers of §2 — but a tens-of-percent perturbation in this thin column, not a dominant boost. (In a larger or denser cell the same physics would matter much more; the conclusion is geometry-specific.)

4 The histories, made visible

The simulation now dumps full per-step neutron trajectories (`--trajdump`). Figure 1 shows real Geant4 histories at thermal, $\sim\text{keV}$ and $\sim\text{MeV}$ over the actual capsule cross-section (beam $\rightarrow +y$), each neutron coloured by the *fraction of its incident energy still remaining*, so the slowing-down reads as a colour change along the path. Red stars mark $^4\text{He}^*$ formation (here the dominant $(n,p)t$; the rare radiative/IPC sub-branch rides on the same captures).

These are exactly the pass-through / scatter-then-leave / capture-at-depth histories that the decade budget of §2 sums over — the displays just make the mechanism legible.

Reproducing. Trajectory dump: `--trajdump` flag (`SteppingAction::DumpTrajectoryStep`); plotter `make_neutron_event_display.py`; build/run recipe in `event_display_RUN.md`. A higher-statistics *E-at-capture vs. E-incident* scatter (how much slowing-down precedes each excitation) is the natural quantitative companion and would come from the full campaign’s terminal data on EOS — noted as a follow-up.

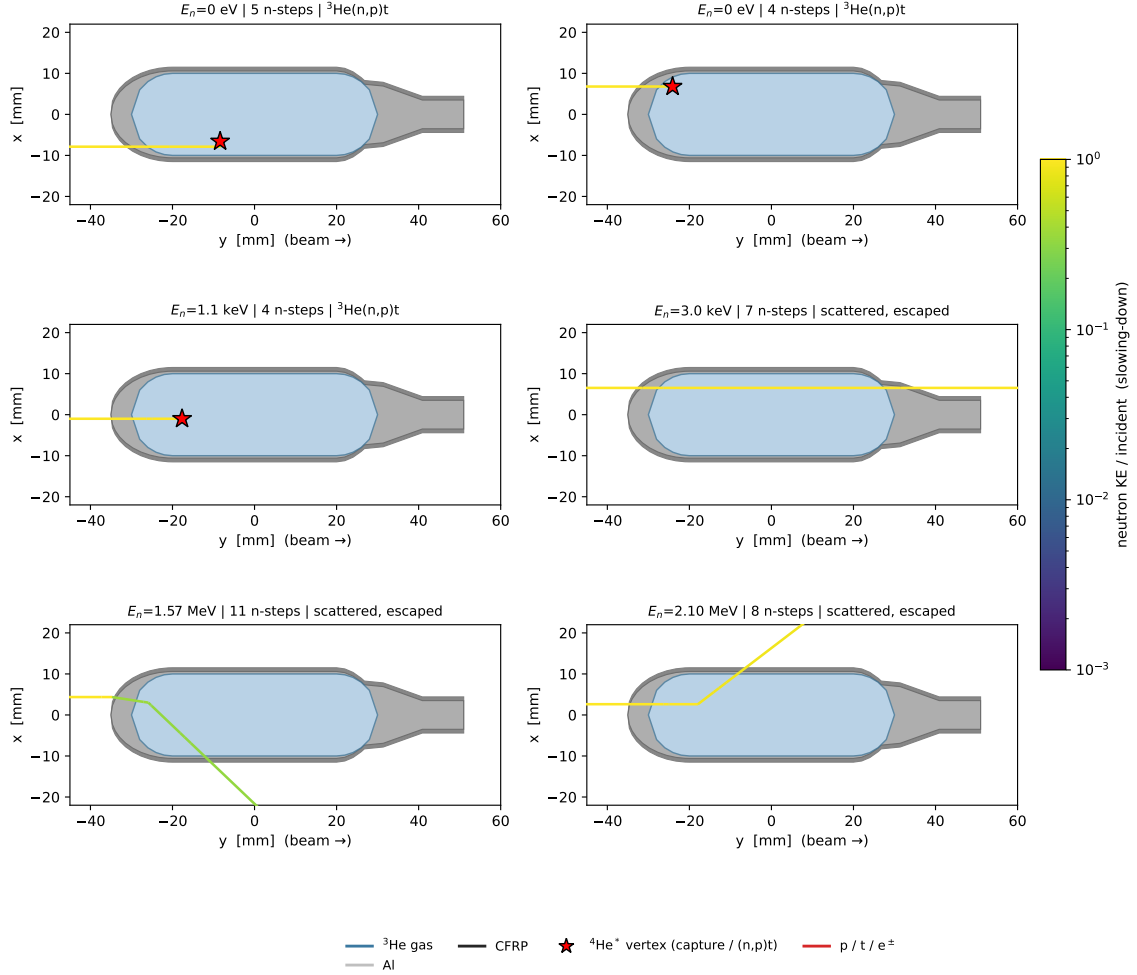


Figure 1: Representative neutron histories in the ^3He capsule, same scattering physics in every panel. **Thermal (top):** absorbed within a fraction of a mm of the upstream gas surface — self-shielding in one picture. **keV (middle):** the neutron traverses a good part of the bore before (n,p)t; captures populate the volume, not just the surface. **MeV (bottom):** the gas is thin so most neutrons cross it, but the elastic histories are rich — the path kinks at each n- ^3He collision and the colour drops from yellow toward green as a single scatter sheds a large chunk of energy. Most escape; the minority that slow down get the “second chance” to capture.